Divide and Conquer Example – Longest Common Prefix

Given n strings, the problem is to find the longest common prefix of these strings. As an example, if the input is {"abdullah", "abdi", "abdal", "abdulkerim"}, output would be "abd". If the input is {"kelam", "kelime", "kemal", "kemik"}, output would be "ke".

(a) Design a divide-and-conquer algorithm for this problem. Please provide a step by step description of your algorithm.

(b) What is the time complexity of your algorithm? Write a recurrence relation and solve it. Provide an answer in terms of n and m, where n is the number of strings and m is the length of the largest string.

LCP (A[0..n-1]) ← string array

if length(A) = 1 return A[0]

Divide all the strings in two parts (left, right)



$T(\frac{n}{2})$ $LCP_1 \leftarrow LCP(A[0..\frac{n}{2}-1])$

$T(\frac{n}{2})$ $LCP_2 \leftarrow LCP(A[\frac{n}{2}..n-1])$

$m \rightarrow$ return the LCP of LCP_1 & LCP_2.

We need compare all the chars one by one, until a mismatch occurs. Return the matching part.

$$T(n) = 2T\left(\frac{n}{2}\right) + m \qquad T(1) = 0$$

$$T(2^k) = 2T(2^{k-1}) + m$$

$$= 2(2T(2^{k-2}) + m) + m = 2^2 T(2^{k-2}) + 2m + m$$

$$= 2^3 T(2^{k-3}) + 2^2 m + 2m + m$$

$$\vdots$$

$$= 2^k \underbrace{T(2^{k-k})}_{0} + 2^{k-1} m + 2^{k-2} m + \dots + m$$

$$= m\left(2^k - 1\right) = m(n-1) \in O(mn)$$

$$\Downarrow \quad \wedge$$